A MORPHISM OF ALGEBRAIC SPACES IS FLAT IF AND ONLY IF IT IS FLAT AT EVERY POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a definition chase; the two cited definitions of this chapter are exactly the ones the statement compares.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The definition of flatness for a morphism of algebraic spaces via a presentation. Definition definition-P, used at proof lines 6020-6021. Equates flatness of f with flatness of the scheme morphism h.
- The definition of flatness at a point for a morphism of algebraic spaces. Definition definition-P-at-point, used at proof lines 6022-6024. Equates flatness of f at x with flatness of h at a point u over x.
- Flatness of a morphism of schemes is flatness at all points of the source. Morphisms, Definition morphisms-definition-flat, used at proof lines 6025-6027. Is the scheme statement the two equivalences reduce to.

Source reference. Original definition 04RD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6181–6181. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 04RE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6182–6182. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 01U3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2396–2396. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Then f is flat if and only if f is flat at all points of $|X|$.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

where U and V are schemes, the vertical arrows are étale, and a is surjective. By definition f is flat if and only if h is flat (Definition 04RD). By definition f is flat at $x \in |X|$ if and only if h is flat at some (equivalently any) $u \in U$ which maps to x (Definition 04RE). Thus the lemma follows from the fact that a morphism of schemes is flat if and only if it is flat at all points of the source (Morphisms, Definition 01U3). $\square$